

DeepGuard: Real-Time Deepfake Detection

Adaptive Multimodal Analysis Architecture

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Table of Contents

1. Problem Statement
2. Existing Solutions
3. Our Approach
4. Model Architecture Design
5. Workflow & Orchestration
6. Frontend App Mechanics
7. RL-Based Fusion Strategy
8. Adaptive Learning & Data Enhancement

Problem Statement

- **The Threat of Deepfakes:** Rapid advancements in generative AI have democratized the creation of hyper-realistic fake audio and video generation.
- **Real-Time Vulnerability:** Scammers are increasingly using deepfakes for real-time video calls, unauthorized facial authentication, and social engineering.
- **The Gap in Defense:** Users lack a transparent, continuous, and highly accurate defense mechanism that can operate *while* they are consuming media on their mobile screens.

Limitations of Existing Solutions

- **Post-Processing Restraints:** Most available tools require users to upload a recorded video file *after* the interaction has ended, making them ineffective for live scams.
- **High Latency Algorithms:** Traditional cloud-based frame analysis suffers from massive network bottlenecks due to naive frame offloading.
- **Single-Modality Bias:** Current detectors usually focus on just video (faces) *or* just audio, creating blind spots against complex attacks.
- **Lack of Protective UX/UI:** Users do not have "always-on" shields that actively scan screen activity without interrupting their daily mobile workflow.

Our Solution: DeepGuard Architecture

- **Real-Time Floating Overlay:** A background Android service that proactively scans the screen via the MediaProjection API while the user runs other applications.
- **Multimodal Fusion:** Concurrent analysis of both Video anomalies (ViTs) and Audio anomalies (Wav2Vec2) to catch multidimensional attacks.
- **Intelligent Smart-Buffer:** Dramatically reduces latency by exclusively subsampling and streaming frames to the server *only when a person is detected* using lightweight edge heuristics.
- **Continuous Active Learning:** A participatory system that learns from user-verified submissions accompanied by an RL-fusion subsystem.

Deep Learning Model Architecture

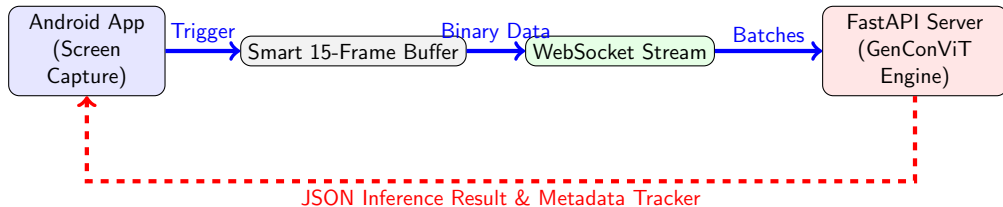
- **Video Inference Core (GenConViT):**

- A powerful ensemble utilizing ConvNeXt-Tiny and Swin Transformers.
- Preceded by an essential 3-tier OpenCV hierarchy (Haar Face → Haar Upper Body → HOG Person Full) filtering out background noise and false positives *before* GPU inference.

- **Audio Inference Core (Wav2Vec2):**

- A transformer fine-tuned specifically for AI-generated speech detection.
- Capable of differentiating authentic human vocal cord acoustics from synthesized text-to-speech architectures (e.g., ElevenLabs, Kokoro).

End-to-End System Workflow

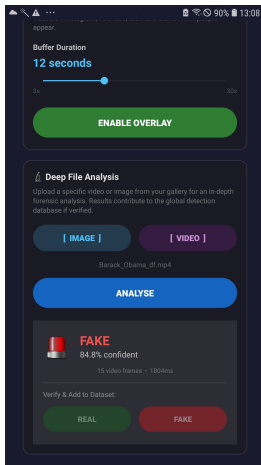


- **Idle Phase:** App runs lightweight `has_person()` checks every 2.0s locally.
- **Stream Phase:** On detection, it grabs exactly 15 subsampled frames over a rolling 3s window and dispatches them as binary bytes over WebSockets.

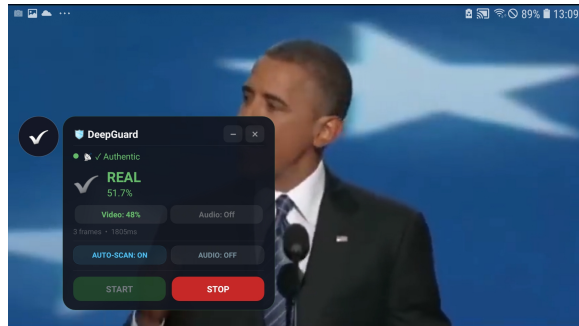
Frontend App Mechanics & Networking

- **Permission Handling:** Secures strict `RECORD_AUDIO` and `SYSTEM_ALERT_WINDOW` (Overlay) privileges reliably.
- **MediaProjection Integration:** Binds directly to the Android internal frame buffer to mirror the screen invisibly without lag.
- **CaptureService State Machine:** Completely segregates operational loads into `IDLE`, `SCANNING`, and `STREAMING` avoiding unnecessary battery drain.
- **Connectivity:** Utilizes persistent `OkHttp` `WebSockets` optimized with `ngrok-skip-browser-warning` headers to bypass standard `HTTP` hurdles during remote deployments.

Frontend User Interface Integration



Deep Analysis Dashboard



Dynamic Floating Overlay

Reinforcement Learning (RL) Fusion Matrix

- **The Core Problem:** How do we mathematically merge Video confidence and Audio confidence effectively when one might be spoofed?
- **Dynamic Modality Weighting:** Unlike a static 50/50 fusion, DeepGuard incorporates an RL mechanism approximating a Multi-Armed Bandit strategy.
- **Update Cycle:** The server evaluates historical stability. If the audio model predicts with consistently sharp margins, its coefficient (weight multiplier) grows dynamically.
- **Result:** If an attacker employs perfect CGI visuals but flawed cloned audio, the RL system penalizes the final merged score drastically due to the audio's heavily weighted failure.

Adaptive Learning & Batch Maintenance

- **Dataset Contribution Mechanics:** Users interacting with the "Deep File Analysis" UI can flag False-Positives via the *Mark Real* or *Mark Fake* dataset buttons.
- **Zero-Trust Storage:** These explicitly verified edge-case files are logged safely inside the verified/real/ and verified/fake/ local server hierarchies (excluded from public VCS).
- **Periodic Batch Training:** A cyclical training routine can then construct micro-batches out of this organic repository, performing localized backpropagation on the GenConViT encoder to adapt the decision boundaries continuously.

Thank You!

Questions, Feedback, and Open Discussion.